

Explainable Deep Learning for Gravitational Lens Detection in Astronomical Survey Images

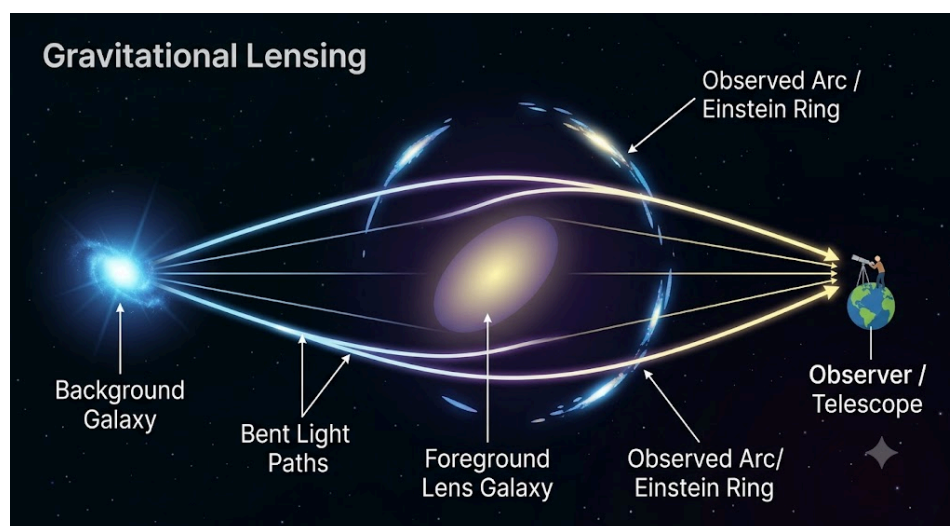
Students: Ahana Bhattacharji and Vaishnav Malvankar

1. Abstract

This project aims to develop a prototype explaining a deep learning system for detecting gravitational lens candidates in astronomical survey images. Strong gravitational lensing occurs when a massive foreground object bends and magnifies light from a distant background source, often producing visible features such as arcs, rings, or multiple distorted images. Since modern astronomical surveys produce very large image datasets and gravitational lenses are rare, manual detection is difficult and does not scale. The project will treat lens detection as an imbalanced image-classification problem. A baseline Convolutional Neural Network will be trained to classify image cutouts as lens or non-lens candidates, with optional comparison to a transfer learning model. Grad-CAM will be used to visualise which image regions influence model predictions and to support false-positive analysis.

2. Introduction

Gravitational lensing occurs when the gravity of a massive object, such as a galaxy or galaxy cluster, bends light from a more distant background source. In strong gravitational lensing, this can create visible distortions such as arcs, rings, or multiple images of the same background galaxy. These systems are scientifically valuable because they help astronomers study distant galaxies, dark matter, and the mass structure of foreground galaxies and galaxy clusters.



In this project, gravitational lensing is treated mainly as an astronomical image-classification problem. The aim is not to model the full physics of lensing, but to build a deep learning pipeline that can identify lens-like visual patterns in survey image cutouts.

3. Motivation

Modern astronomical surveys observe wide regions of the sky and capture millions or billions of celestial objects. This creates an opportunity to discover more gravitational lenses, but it also creates a major data-processing problem. Since strong gravitational lenses are rare, only a very small fraction of survey images are expected to contain real lens candidates.

Manual inspection is not practical at this scale. Although expert astronomers can identify lens-like features, checking large survey datasets by hand would be slow and inconsistent. Automated machine learning methods can help by filtering large image collections and prioritising likely candidates for human review.

The task is also difficult because many non-lens objects can resemble lensing features. Spiral galaxies, ring galaxies, galaxy mergers, overlapping objects, bright stars, and imaging artefacts can appear similar to arcs or rings. This makes false positives a major issue. Therefore, this project focuses not only on classification, but also on imbalance-aware evaluation, false-positive analysis, and explainability using Grad-CAM.

4. Current Line of Thought

Previous work has shown that CNN-based methods can be useful for strong gravitational lens detection. Lanusse et al. (2017) demonstrated early deep learning methods for automatic galaxy-galaxy strong lens detection. Metcalf et al. (2019) and Bom et al. (2022) framed lens detection as a machine learning challenge and highlighted the importance of controlling false positives in rare-object detection. Pearce-Casey et al. (2024) further showed that models trained on controlled or simulated data can be harder to apply to real survey images. For explainability, Grad-CAM, introduced by Selvaraju et al. (2017), provides a way to visualise which image regions influence CNN predictions.

4.1 Data Collection

The project will use or collect a labelled dataset containing gravitational lens and non-lens image cutouts. Each image should have a class label indicating whether it contains a lens candidate. The dataset may come from public lens-finding challenges, existing survey image

datasets, or simulated lens datasets. If simulated data is used, tools such as *lenstronomy* may be relevant, although the limitations of simulated data compared with real survey images will need to be discussed.

4.2 Data Cleaning and EDA

The images will be inspected and preprocessed before model training. This may include resizing images to a fixed input size, normalising pixel values, removing unusable or corrupted samples, and splitting the data into training, validation, and test sets. Exploratory Data Analysis will include checking class balance, viewing example images, and identifying possible noise, artefacts, or visually confusing non-lens cases. Since lenses are rare, class imbalance is expected to be an important issue.

4.3 Model Training

A baseline Convolutional Neural Network will be trained to classify image cutouts as lens or non-lens candidates. If time allows, this model may be compared with a transfer learning model such as ResNet, EfficientNet, MobileNet, or DenseNet. Because accuracy can be misleading for imbalanced datasets, evaluation will use precision, recall, F1-score, confusion matrix analysis, ROC-AUC, and PR-AUC. False-positive examples will also be analysed to understand which non-lens objects are most commonly misclassified.

Grad-CAM will be used to visualise model attention for selected predictions. These heatmaps will help assess whether the model focuses on meaningful lens-like structures such as arcs and rings, or whether it is misled by irrelevant features such as spiral arms, bright stars, overlapping objects, noise, or imaging artefacts.

5. Expected Output

The expected output is a student-scale prototype explainable deep learning system for gravitational lens candidate detection. The system will classify astronomical image cutouts as lens or non-lens candidates, evaluate model performance using imbalance-aware metrics, analyse false positives, and generate Grad-CAM heatmaps for selected predictions. The final system is not intended to be a production survey pipeline, but a focused prototype demonstrating how deep learning and explainable AI can support gravitational lens detection.

6. References

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